

National University of Computer and Emerging Sciences



Lab Manual 04 Fundamentals of Big Data Lab

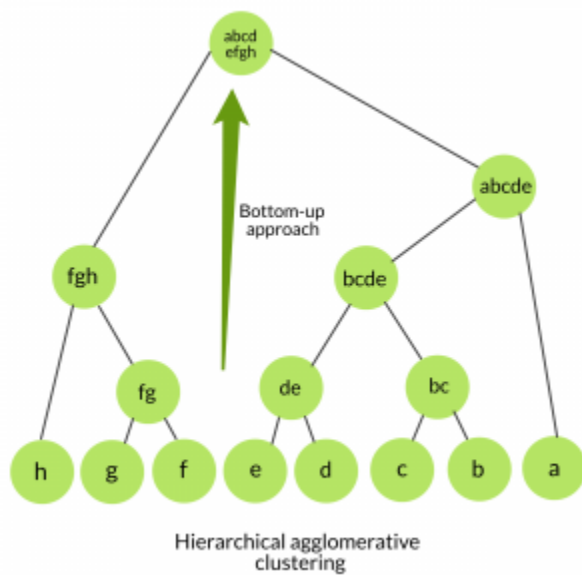
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Section	
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In data mining and statistics, hierarchical clustering analysis is a method of cluster analysis that seeks to build a hierarchy of clusters i.e. tree-type structure based on the hierarchy.

Agglomerative Clustering: Also known as bottom-up approach or hierarchical agglomerative clustering (HAC). A structure that is more informative than the unstructured set of clusters returned by flat clustering. This clustering algorithm does not require us to prespecify the number of clusters. Bottom-up algorithms treat each data as a singleton cluster at the outset and then successively agglomerates pairs of clusters until all clusters have been merged into a single cluster that contains all data.

Algorithm :

```
given a dataset ( $d_1, d_2, d_3, \dots, d_N$ ) of size N
# compute the distance matrix
for i=1 to N:
    # as the distance matrix is symmetric about
    # the primary diagonal so we compute only lower
    # part of the primary diagonal
    for j=1 to i:
         $dis\_mat[i][j] = distance[d_i, d_j]$ 
each data point is a singleton cluster
repeat
    merge the two cluster having minimum distance
    update the distance matrix
until only a single cluster remains
```



Python implementation of the above algorithm using the scikit-learn library:

- Python3

```
from sklearn.cluster import
AgglomerativeClustering
import numpy as np

# randomly chosen dataset
X = np.array([[1, 2], [1, 4], [1, 0],
              [4, 2], [4, 4], [4, 0]])

# here we need to mention the number of clusters
# otherwise the result will be a single cluster
# containing all the data
clustering = AgglomerativeClustering(n_clusters =
2).fit(X)

# print the class labels
print(clustering.labels_)
```

Output :

```
[1, 1, 1, 0, 0, 0]
```

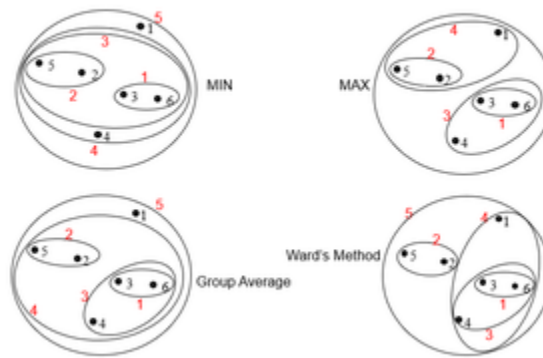
Computing Distance Matrix: While merging two clusters we check the distance between two every pair of clusters and merge the pair with least distance/most similarity. But the question is how is that distance determined. There are different ways of defining Inter Cluster distance/similarity. Some of them are:

1. **Min Distance:** Find minimum distance between any two points of the cluster.
2. **Max Distance:** Find maximum distance between any two points of the cluster.
3. **Group Average:** Find average of distance between every two points of the clusters.

4. Ward's Method: Similarity of two clusters is based on the increase in squared error when two clusters are merged.

For example, if we group a given data using different method, we may get different results:

Hierarchical Clustering: Comparison



Task A:

1. Use built in Agglomerative to work on given data, use annual income and spending score.

Task B:

1. Design your own Agglomerative clustering algorithm, use annual income and spending score.